

회귀분석

정재환

Data science summer institute

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Introduction

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Agenda

1. What is regression analysis?
2. Steps in regression analysis

Scientific method

A scientific method is the application of logic and objectivity to the understanding of phenomena in order to predict future events.

A characteristic feature of the scientific method is its cyclic nature since the facts ending one cycle are the beginning of the next.

Induction refers to formulation of a theory to explain the observed facts (data); data $\Rightarrow$ theory

Deduction consists of logical analysis of what a general theory law says about a particular event; theory $\Rightarrow$ data

Statistics

Statistics is the *data science* where data are numbers with a context.

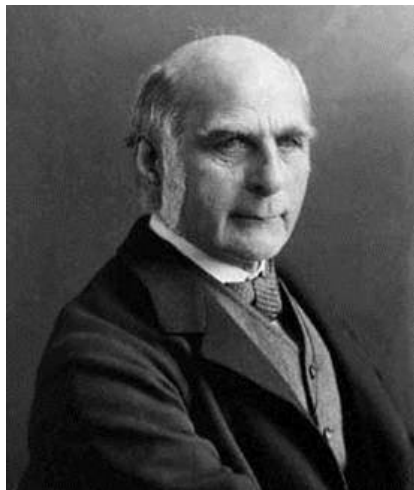
The field of science concerned with the collection and analysis of numerical information (*data*) in order to answer questions wisely (Brillinger).

Statistics adopts an inductive reasoning whereas mathematics (probability theory), an deductive reasoning.

회귀 의미

- ▶ 회귀, regression, 回歸
- ▶ 회귀라는 말은 이전 발전단계로 돌아감 의미 (네이버)
- ▶ 통계학에서는 평균 **mean** 으로 돌아감
- ▶ 개념상 변수들간 함수관계를 탐색하는 방법
 - ▶ 반응변수 **response variable**: 출력변수 **output**, target variable, dependent variable
 - ▶ 설명변수 **explanatory variable**: 입력변수 **input**, 예측변수 **predictor**, covariate, regressor, factor, carrier, independent variable
- ▶ 응용
 - ▶ economics, finance, law, meteorology, medicine, biology, chemistry, engineering physics, education, sports, history, sociology, psychology

Francis Galton



- ▶ Sir Francis Galton, England, 1822-1911
- ▶ Inventor of regression[Galton, 1886]
- ▶ 가족간 유사성^{resemblance}에 관심을 가짐

입출력관계 모형화

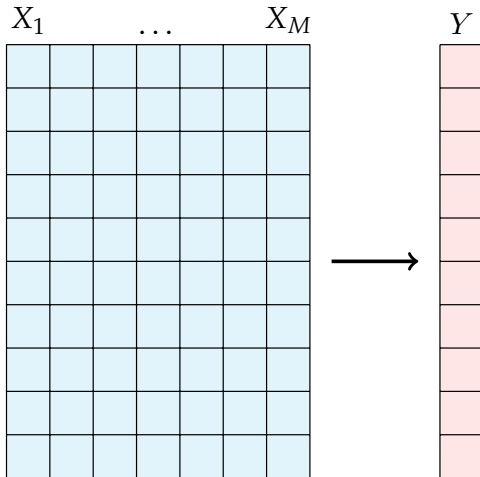
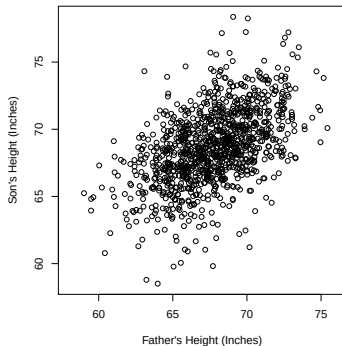
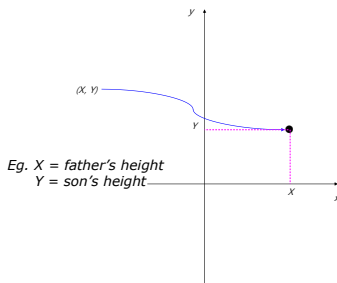


Figure: $X_1, \dots, X_M$ 들은 설명변수들을, Y 는 반응변수를 나타냄

산점도

- ▶ 부모의 키(설명변수)와 자식의 키(반응변수) 간의 관계를 연구하고자 1078명 부모 및 자식의 키를 관측¹



¹[Pearson and Lee, 1903]

히스토그램

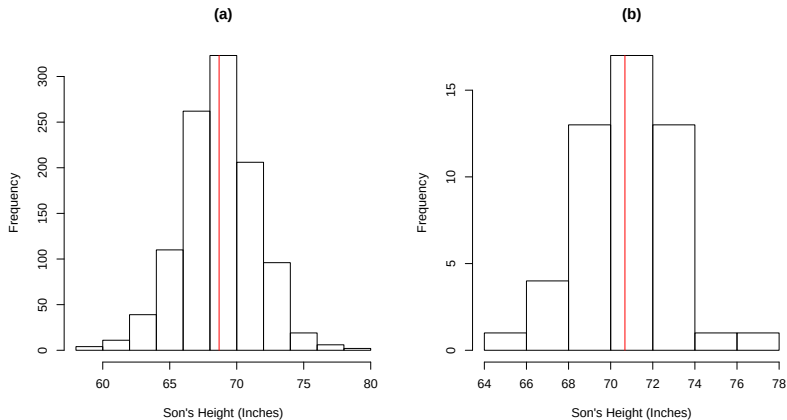
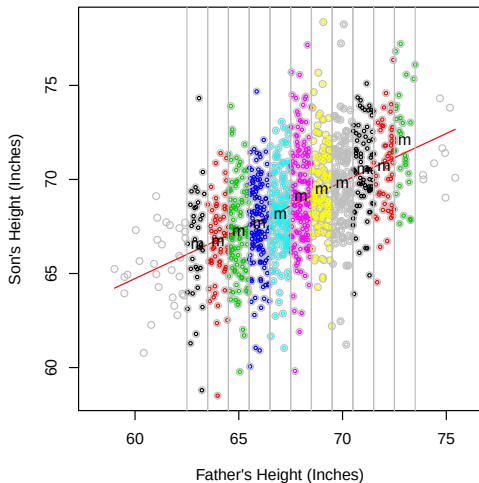


Figure: (a) 전체 자식들에 대한 히스토그램; (b) 부모의 키가 72인치인 자식들에 대한 히스토그램.

히스토그램과 평균 예측

- ▶ 랜덤하게 선택된 자식의 키를 어떤 값으로 예측?
 - ▶ 자식들의 키의 평균이 69 인치 (그림 (a)의 붉은선)
- ▶ 부모의 키가 72인치라는 조건하에서 랜덤하게 선택된 자식의 키를 어떤 값으로 예측?
 - ▶ 그룹 자식들의 키의 평균이 71 인치 (그림 (b)의 붉은선)

국소평균과 회귀 직선



골턴 데이터에 대한 단순선형모형 적합

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	33.89	1.83	18.49	0.00
x	0.51	0.03	19.01	0.00

Table: 골턴 데이터에 대한 선형모형적합 결과

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
x	1.00	2144.58	2144.58	361.23	0.00
Residuals	1076.00	6388.00	5.94		

Table: 골턴 데이터에 대한 선형모형적합에 대한 분산분석

Definition of Regression

Regression is the *conditional expectation (mean)* of the response Y given the predictor $X = x$.

For bivariate normal case, the regression is given by a *line*.

Regression Model

f describes the relationship between the response Y and the predictors $X_1, X_2, \dots, X_p$.

Notations

- ▶ $Y = f(X_1, X_2, \dots, X_p) + \varepsilon$
- ▶ Y : response variable
- ▶ $X_1, X_2, \dots, X_p$: predictors
- ▶ ε : random error

$f(x)$ = conditional mean of the response Y given the predictor values x

Linear Regression Model

Definition:

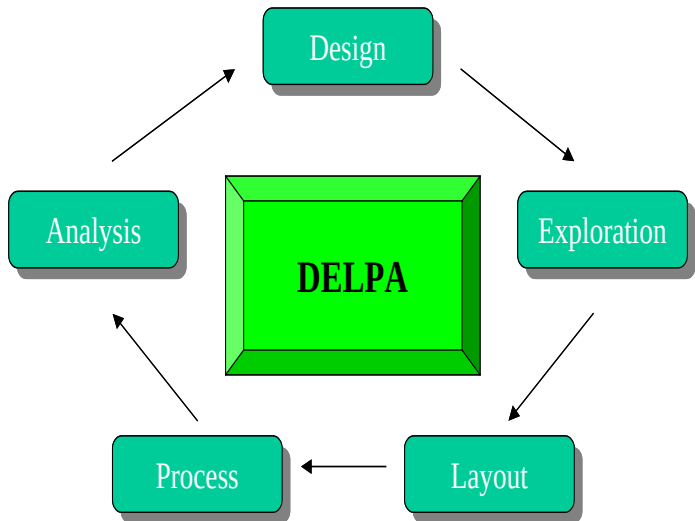
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

$\beta_0, \beta_1, \dots, \beta_p$: regression parameters or coefficients

Parameters are unknown constants to be determined (estimated) from the data.

Steps in Regression Analysis

1. Statement of the problem
2. Selection of potentially relevant variables
3. Data collection
4. Model specification
5. Choice of fitting method
6. Model fitting
7. Model validation and criticism
8. Using the chosen model(s) for the solution of the posed problem



1. Statement of the Problem

This includes the determination of the question to be addressed by the analysis.

It is the first and perhaps the most important step.

An ill-defined problem or a mis-formulated question can lead to wasted effort.

2. Selection of Potentially Relevant Variables

This step is to select a set of variables to explain or predict the response variable.

Example:

- ▶ Response: price of a single family house
- ▶ Predictors: area of the house, age of the house, number of bedrooms, etc.

3. Data Collection

This step is to collect the data from the environment under study to be used in the analysis.

n : sample size

column: a variable

row: an observation

Data Structure

Observation Number	Response Y	Predictors			
		X_1	X_2	$\cdots$	X_p
1	y_1	x_{11}	x_{12}	$\cdots$	x_{1p}
2	y_2	x_{21}	x_{22}	$\cdots$	x_{2p}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	y_n	x_{n1}	x_{n2}	$\cdots$	x_{np}

Types of Variables

Types:

- ▶ quantitative
 - ▶ house prices, number of bedrooms, age, taxes
- ▶ qualitative
 - ▶ credit type (good or bad), gender (male or female)
 - ▶ coded as indicator or dummy variables

Types of Variables in Regression Analysis:

- ▶ Response: quantitative
- ▶ Predictors: quantitative/qualitative

Special Names

Analysis of variance

- ▶ Response: quantitative
- ▶ Predictors: qualitative

Analysis of covariance

- ▶ Response: quantitative
- ▶ Predictors: quantitative/qualitative

Logistic regression

- ▶ Response: qualitative (usually binary)
- ▶ Predictors: quantitative/qualitative

4. Model Specification

Initial model can be specified by the experts.

The hypothesized model can then be either confirmed or refuted by the analysis of the collected data.

Classification of Models

Linear vs nonlinear: parameters enter into the mean of the response linearly (nonlinearly).

Simple (one predictor) vs multiple (more than one predictor)

Univariate (one response) vs multivariate (more than one response)

Examples

1. $Y = \beta_0 + \beta_1 X_1 + \varepsilon$
 2. $Y = \beta_0 + e^{\beta_1 X_1} + \varepsilon$
 3. $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$
 4. $Y_1 = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$ and $Y_2 = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$
 5. $Y = \beta_0 + \beta_1 \log X_1 + \varepsilon$
 6. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$
- ▶ Linear: 1, 3, 4, 5, 6
 - ▶ Nonlinear: 2
 - ▶ Simple: 1, 2, 5
 - ▶ Multiple: 3, 4, 6
 - ▶ Univariate: 1, 2, 3, 5, 6
 - ▶ Multivariate: 4

Why do we care for linearity?

If a model is linear, then one can use an ordinary regression software.
Same software can be used to estimate parameters for models 1, 3, 4, 5, 6.

5. Method of Fitting

It is to estimate the parameters of the model.

One of the most commonly used methods of estimation is the least squares method.

Other methods include the maximum likelihood method, the ridge method, and the principal components method.

6. Model Fitting

It is to estimate the regression parameters or to fit the model to the collected data.

Denote the predicted value by

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \cdots + \hat{\beta}_p X_p,$$

where $\hat{\beta}_0, \hat{\beta}_1, \cdots, \hat{\beta}_p$ are estimators of regression parameters.

The fitted values are given by

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}, \quad i = 1, \cdots, n.$$

The difference between fitted and predicted values is that the fitted value refers to the case where the values used for the predictor variables correspond to one of the n observations in our data, but the predicted values are obtained for any set of values of the predictor variables.

Objectives of Regression Analysis

Explicit determination of the regression equation

A summary of the relationship between the response and the predictors

Usage

- ▶ to evaluate the importance of individual predictors
- ▶ to forecast values of the response for a given set of predictors

References



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